
CSCI-121 · Week 2, Day 1

Name _____ Section (circle): R3 Mon / R4 Wed

The two rules from today

Java widens for free. Java **never** narrows without a cast.

`==` for primitives. `.equals()` for everything else.

The eight primitives

Everything else in Java is a reference.

`byte` `short` `int` `long` `float` `double` `char` `boolean`

Lowercase, always. **If the type starts with a capital letter, the variable holds an arrow.**

Predict, then we run it

Write your answer *before* it goes on the screen. Being wrong here is free; being wrong in three weeks is not.

	Code	Your prediction
1	<code>System.out.println(7 / 2);</code>	
2	<code>System.out.println(7 / 2.0);</code>	
3	<code>int x = 3.9;</code>	
4	<code>int x = (int) 3.9;</code>	
5	<code>char c = 'A'; System.out.println(c + 1);</code>	
6	<code>System.out.println("" + 1 + 2);</code>	
7	<code>System.out.println(1 + 2 + "");</code>	

Draw the boxes

```
int n = 5;
String s = "hi";
```

Draw what `n` holds and what `s` holds. One of them has an arrow in it.

The one that matters

```
int[] x = {1, 2, 3};
int[] y = x;
y[0] = 99;
System.out.println(x[0]);
```

Prediction: _____

What actually happened, and why — one sentence:

== and .equals()

```
String a = "hi";
String b = new String("hi");
String c = "hi";
```

	Answer	Why
a == b		
a.equals(b)		
a == c		

The last row is the dangerous one. In one sentence, why is it dangerous to rely on?

This week

Q1 — Tuesday Sep 8 , 10 min, start of class, on paper, no notes	data types, operators, strings
Recitation 2 — finish at home, push, open the PR	Sun Sep 6, 11:59 PM
Drills — Asterisks · Show me the Numbers	Sun Sep 6, 11:59 PM
Skill Builder 1	Sun Sep 13, 11:59 PM
Reading — Think Java Ch 2, Ch 3, §6.10, §9.3	nothing tracks it; look things up

Everything on Q1 is in Recitation 2 — and it is due two days before the quiz. If your tests are green, you have studied.

Exit ticket — tear this off and hand it in at the door

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Keep the other pages. This one is mine.

1. `int x = 7 / 2;` — what is `x`?

2. Draw the box for `int n = 5;` and the box for `String s = "hi";`

3. Two strings print the same text. `==` says `false`. In one sentence, why?
